

Generation Interconnection System Impact Study Report

For Queue Project Q21-1826

International Paper Valliant Ok 345 kV

September 18, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1826. The Interconnection Customer proposes a 1411.7 MW natural gas combined cycle facility interconnecting at the International Paper Valliant Ok 345 kV substation in the Virginia area. The request was received on July 22, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 604 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$58,771,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CLARK COUNTY - SHELTON 138.0 kV	ARTESIA COUNTRY CLUB - CLARK COUNTY 345.0 kV	187.5	246	8.5
SHELTON SUBSTATION - FIVE CHANNELS 138.0 kV line	ARTESIA CUNTRY CLUB - CLARK COUNTY 345.0 kV Line	182.7	239	8.5
KITIMAT - TERROR LAKE MICROGRID 138.0 kV	INTERNTIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV	171.6	283	5.9
KITIMAT - CASCADE STEEL 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV Line	168.9	296	5.9
FEAGAN'S MILL - CASCADE STEEL 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	165.5	310	5.9
COW - FEAGAN'S MILL 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	164.9	315	5.9
COW SUBSTATION - YAMHILL 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	162.6	328	5.9
3 MILE KNOLL - YAMHILL 138.0 kV line	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV L	161.4	335	5.9
EAST ELLENSBURG - WEST HILLS 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV L	159.5	282	5.9
PALMYRA - GRIMMWAY-MALAGA 138.0 kV	CLARK COUNTY - KEARNEYVILLE 345.0 kV L	156.8	313	6.0
EAST ELLENSBURG - TERROR LAKE MICROGRID SUBSTATION 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	156.8	296	5.9
PALMYRA - GLENBROOK 138.0 kV	CLARK COUNTY - KEARNEYVILLE 345.0 kV Line	155.0	325	6.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GLENBROOK - REDMAN 138.0 kV	CLARK COUNTY - KEARNEYVILLE 345.0 kV	153.5	336	6.0
GALLEGOS - REDMAN 138.0 kV	CLARK COUNTY - KEARNEYVILLE 345.0 kV Line	152.6	344	6.0
KILLINGSWORTH - GALLEGOS 138.0 kV	CLARK COUNTY - KEARNEYVILLE 345.0 kV	151.3	354	6.0
LCEC NO. 5 - SHEPHERDSTOWN 138.0 kV line	KEARNEYVILLE SUBSTATION - CLARK COUNY SUBSTATION 345.0 kV	149.8	338	10.9
SANTA ROSA ENERGY CTR - MONOLITH 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV	144.1	351	8.0
CELERON - SOUTHAMPTON 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV Line	139.0	294	7.9
MABTON SUBSTATION - SOUTHAMPTON 138.0 kV	KEARNEYVILLE SUBSTATION - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	136.9	306	7.9
PLUM POINT ENERGY STA - MABTON 138.0 kV Line	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV	135.5	315	7.9
INTERNATIONAL PAPER VALLIANT OK 345 kV/138 kV Transformer Bank	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV	133.6	499	16.9
BELLEAIR - WPCP 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK SUBSTATION 345.0 kV L	133.5	371	5.7
UNIVERSITY OF MAINE - PLUM PT ENERGY STATION 138.0 kV	KEARNEYVILLE - INTERNATIONAL PAPER VALLIANT OK 345.0 kV	132.7	334	7.9
UNIVERSITY OF MAINE - THREE PEAKS 138.0 kV	INTERNATIONAL PAPER VALLIANT OK - KEARNEYVILLE 345.0 kV Line	131.5	343	7.9
THREE PEAKS - OCEAN RD 138.0 kV Line	INTERNATIONAL PAPER VALLIANT OK - KERNEYVILLE 345.0 kV	130.9	348	7.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CLARK COUNTY - SHELTON 138.0 kV	Rebuild line with higher-capacity conductor	\$1,522,000	72% (\$1,090,000)
FIVE CHANNELS - SHELTON 138.0 kV	Rebuild line with higher-capacity conductor	\$1,670,000	72% (\$1,196,000)
KITIMAT - TERROR LAKE MICROGRID 138.0 kV	Rebuild line with higher-capacity conductor	\$4,091,000	68% (\$2,774,000)
KITIMAT - CASCADE STEEL 138.0 kV	Rebuild line with higher-capacity conductor	\$2,499,000	68% (\$1,694,000)
FEAGAN'S MILL - CASCADE STEEL 138.0 kV	Rebuild line with higher-capacity conductor	\$2,835,000	68% (\$1,922,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COW - FEAGAN'S MILL 138.0 kV	Rebuild line with higher-capacity conductor	\$2,042,000	68% (\$1,384,000)
YAMHILL - COW 138.0 kV	Rebuild line with higher-capacity conductor	\$3,204,000	68% (\$2,172,000)
3 MILE KNOLL - YAMHILL 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,183,000	68% (\$2,836,000)
EAST ELLENSBURG - WEST HILLS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,795,000	68% (\$1,895,000)
PALMYRA - GRIMMWAY-MALAGA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,161,000	74% (\$1,597,000)
EAST ELLENSBURG - TERROR LAKE MICROGRID SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,531,000	68% (\$1,716,000)
GLENBROOK - PALMYRA 138.0 kV	Rebuild line with higher-capacity conductor	\$1,588,000	74% (\$1,174,000)
REDMAN SUBSTATION - GLENBROOK SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,690,000	74% (\$1,249,000)
GALLEGOS - REDMAN 138.0 kV	Rebuild line with higher-capacity conductor	\$1,662,000	74% (\$1,229,000)
GALLEGOS - KILLINGSORTH 138.0 kV	Rebuild line with higher-capacity conductor	\$3,479,000	74% (\$2,572,000)
LCEC NO. 5 - SHEPHERDSTOWN 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,170,000	63% (\$2,007,000)
SANTA ROSA ENERGY CTR - MONOLITH 138.0 kV	Rebuild line with higher-capacity conductor	\$4,368,000	70% (\$3,041,000)
SOUTHAMPTON - CELERON 138.0 kV	Rebuild line with higher-capacity conductor	\$1,888,000	70% (\$1,314,000)
MABTON SUBSTATION - SOUTHAMPTON 138.0 kV	Rebuild line with higher-capacity conductor	\$2,520,000	70% (\$1,754,000)
PLUM POINT ENERGY STA - MABTON 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,808,000	70% (\$1,954,000)
INTERNATIONAL PAPER VALLIANT OK 345/138 kV Auto	Replace transformer with higher-rated unit	\$20,185,000	69% (\$14,013,000)
BELLEAIR - WPCP 138.0 kV	Rebuild line with higher-capacity conductor	\$2,569,000	74% (\$1,900,000)
UNIVERSITY OF MAINE - PLUM PT ENERGY STATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,249,000	70% (\$869,000)
UNIVERSITY OF MAINE - THREE PEAKS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,292,000	70% (\$1,596,000)
THREE PEAKS - OCEAN RD 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,493,000	70% (\$3,823,000)
	Total allocated to project		\$58,771,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.